

Executive Analytical Report: Collections Performance & Strategy Audit

This report addresses the leadership team's concerns regarding the reported 11% month-over-month (MoM) improvement in collections recovery. By auditing approximately 12 months of collections data across multiple operational channels (including human calling, progressive dialing, IVR, WhatsApp, SMS, and field visits), we have reconstructed actual business performance to determine where the next ₹10 Cr investment should be deployed.

1. Data Quality & Forensics (The "Golden Dataset" Pipeline)

The foundation of this analysis required untangling a complex web of raw data spanning 17 relational datasets. The raw data was highly fragmented, lacking a single source of truth and containing intentional systemic issues. To build a reliable "Golden Dataset," we detected and treated several critical data traps:

- **Phantom Recovery (Duplicate Payments):** We identified duplicate payment events and duplicate payment references that were artificially inflating gross recovery figures. These were strictly deduplicated to extract the true chronological payment events.
- **Timezone Skew:** Event timestamps were recorded in mixed, localized timezones, specifically UTC, Asia/Kolkata, and Asia/Dubai. Failing to normalize these would corrupt any "calling time" driver analysis. All temporal data was standardized to UTC.
- **The Borrower ID Trap & Identity Fragmentation:** We uncovered multiple identifiers for the same agent and inconsistent identifiers across the tables. Entity resolution was applied to map events correctly and discard randomized identifiers that would have derailed demographic analyses.
- **Temporal Latency:** We isolated late-arriving events and conflicting timestamps (e.g., overwritten-style status history). The `account_status_history` table was treated as an append-only log, anchoring purely on actual event timestamps rather than delayed system record times.
- **Disposition Code Drift:** The data contained changed schema versions and legacy disposition codes. For instance, overlaps in "Promise to Pay" aliases required harmonizing codes to prevent the silent underreporting of successful conversion metrics.

2. Is the 11% MoM Improvement Real?

Leadership's skepticism regarding the 11% MoM recovery improvement was justified. Our independent metric reconstruction challenged the existing definitions of Contact rate, PTP (Promise to Pay) rate, and Recovery rate.

- **Metric Correction:** The 11% figure was heavily skewed by the ingestion of duplicate payment events and attribution errors (e.g., payments incorrectly attributed to the latest campaign rather than the true causal interaction).
- **True Performance:** After filtering out phantom recoveries and correcting the PTP rate to include all legacy disposition codes, the true operational trend proved to be highly volatile and statistically flat. The 11% spike was an artifact of denominator manipulation and data-quality failures rather than a structural operational improvement.

3. Drivers of Change & Counterfactual Analysis

To understand *why* performance fluctuated, we analyzed underlying portfolio dynamics, including DPD (Days Past Due), portfolio mix, client demographics, and campaign channels.

Leadership introduced new targeting strategies (e.g., moving from "legacy" to "v1/v2/v3" strategies) midway through the year. Because these campaigns ran concurrently, we utilized a counterfactual matching methodology to isolate their true impact.

- **Methodology:** We applied risk-stratified matching to compare identical borrower segments (Treatment vs. Control).
- **Findings:** The targeting mix remained relatively static, ruling out Simpson's paradox or shifting portfolio risk as the primary cause of the flat baseline. However, the isolated matching model proved that specific new targeting strategies generated a measurable incremental lift per account when completely divorced from legacy data leaks.

4. Strategic Recommendation: Deploying the ₹10 Cr Investment

Leadership has restricted the ₹10 Cr investment to one of six areas: better telephony infrastructure, more agents, AI voice automation, better borrower targeting, WhatsApp/digital engagement, or field operations.

Based on the counterfactual analysis, we recommend deploying the ₹10 Cr into **Better Borrower Targeting**.

- **Expected Financial Impact & ROI:** The new targeting strategies (when properly isolated) yield a positive incremental recovery per account.
- **Break-Even Point:** Given the current targeted volume (driven by daily_targeting lists), this investment is projected to break even well within the first 12 months, assuming portfolio volumes remain stable.
- **Downside Scenario:** If the total addressable market of active accounts shrinks, or if borrower fatigue degrades the efficacy of the new targeting models over time, the break-even horizon will naturally extend.

5. Production Analytics Architecture

To ensure leadership has daily, 60-second visibility into actual performance, this analysis must be productionized. We recommend the following data engineering pipeline:

- **Raw to Staging:** Ingest all 17 raw sources without schema enforcement to prevent upstream breakages, followed by UTC timestamp normalization and late-arriving data flags.
- **Clean to Golden:** Apply strict deduplication on primary keys (e.g., payment_id, call_id). Implement Slowly Changing Dimensions (SCD) for agent and borrower histories to prevent attribution window bias.
- **Feature to Metrics:** Pre-calculate challenging metrics (True PTP Rate, Recovery per Account) using explicit data contracts to ensure definitions cannot silently drift.
- **Monitoring & Anomaly Detection:** Deploy automated daily checks to flag duplicate payments or conflicting timestamps before they reach the executive dashboard.

By shifting away from easily manipulated naive metrics and anchoring on this strictly deduplicated golden pipeline, the business can confidently execute its new targeting strategy and track its true return on invested capital.